Mohamed Aarif R

Location: Chennai, India (Open To Relocate)
Email: hi.aarif97@gmail.com **Mobile:** +91-8667882042
Linkedin Profile **GitHub**

PROFESSIONAL SUMMARY

Certified Senior AI Data Engineer with over 7 years of expertise in AI Engineering, Data Engineering, DataOps, and MLOps. Extensive knowledge in the Cybersecurity domain, coupled with a proven track record of designing and deploying robust data and AI platform applications for large-scale knowledge graph management and Big Data workloads. Possesses a cloud-agnostic mindset when crafting solutions, with rich experience in architecting agentic AI systems, multi-agent frameworks, and RAG pipelines that power real-time intelligence and autonomous operations. Demonstrated proficiency in platform optimization, modernization, containerization, and platform reliability engineering aligned with DevSecOps principles. Excels in analytical, problem-solving, and communication skills. Deep hands-on expertise in Azure and GCP, enabling delivery of highly available solutions and establishment of platforms for scaling AI/ML services and full ML lifecycle management. Solid understanding of data security best practices, including data encryption, access controls, and compliance requirements. Well-versed in OLTP, OLAP, and graph data systems, with a commitment to continuous learning and staying ahead of emerging technologies.

WORK EXPERIENCE

Ernst & Young

May 2022 - Present

Role: Senior AI Data Engineer **Location: Chennai, India**

Project Name: Knowledge AI Fabric for Cyber Analytics and Defense (KAFCAD) **Nov 2024- Present**

Project Description: A unified cyber defense and resilience module for CyberOps that harnesses agentic and recursive AI with a knowledge graph lake, providing a 360-degree view of an organization's security posture by unifying hundreds of cyber data feeds into a large-scale Knowledge Graph with billions of traversable data paths. The platform incorporates a semantic enrichment layer powering autonomous multi-agent AI systems for real-time threat intelligence extraction, vulnerability prioritization, and automated security operations across the enterprise.

- Engineered Delta Lake ingestion pipelines on Azure Databricks using a **medallion architecture** to normalize and synchronize intelligence feeds into a unified vulnerability repository, with **incremental ingestion**, control tables, and **automated data-quality** validation.
- Architected **fully automated DataOps workflows** using **Databricks Asset Bundles** (DABs) integrated with Azure DevOps, enabling **end-to-end CI/CD** for Python wheel builds, library management, notebook deployments, and scheduled job orchestration across environment promotions.
- Modeled and deployed a high-performance, enterprise-scale **Knowledge Graph** in Neo4j, enabling complex multi-hop analytics such as CVE-to-Threat Actor attribution; performed entity resolution, **ontology normalization**, and graph migration to maintain integrity across **millions of interconnected entities**.
- Built an automated CTI Knowledge Graph pipeline using **AutoGen**, BAML, and Python with **multi-agent domain-specialist extractor-validator swarm pairs** for schema-validated entity extraction from unstructured data (threat reports), and deduplication; ingested **extracted entities to Neo4j with cosine vector indexes**, and **document chunks with vector embeddings** to Elasticsearch using **langchain**.
- Architected a Knowledge Graph query layer as a **MCP server via FastMCP**, implementing an intent-classified pipeline with AutoGen agents route questions through **semantic (vector search + LLM reranking) or factual (Cypher generation + fallback variations) paths** over Neo4j; built **LangChain-based RAG pipelines** fusing Elasticsearch document context and MCP graph context to power CTI report generation.
- Implemented Docker **multi-stage containerization** and Kubernetes (KaaS) deployments including ConfigMaps, Services, Ingress routing, and Nginx reverse proxy; integrated **Aqua Security container scanning** into the CI/CD pipeline for **vulnerability-free image promotion**.
- Architected an **AIOps observability layer** using the ELK Stack (Elasticsearch, Logstash, Kibana) and Prometheus/Grafana to aggregate logs, **monitor Neo4j query latency, API performance, and pipeline SLAs**; implemented intelligent alerting rules and automated anomaly detection for pipeline failures, **agent latency spikes**, and cluster resource thresholds.

Project Name: CyberOps Data Platform **May 2022 - Present**

Project Description: Cyber Operations suite of products is aimed to help accelerate, sustain cyber security operation capabilities focusing on AI powered advanced visualization & reporting, data analytics & data repository, orchestration and automation.

- Led the design and implementation of a **scalable**, **cloud-agnostic**, **multi-tenant DataOps architecture**, integrating diverse **50+ cyber data sources** into a comprehensive relational and graph data models adhering to strict data governance principles to enhance integrity and security.
- Contributed to cyber R&D activities, product scope discussions, cost-benefit analysis, strongly advocated the benefits of supporting **dual implementation pattern (SaaS, self-hosted),** and shaping go-to-market strategy, which resulted in a **45%** increase in client adoption and accelerated onboarding of high-profile clients.
- Created and optimized reusable **Python and PySpark utilities** for data extraction from diverse sources (REST APIs, GraphQL, data lakes, data warehouses, data estates, databases, files, SaaS, event streams) promoting **code reusability** and reducing development effort by **30%.**
- Designed and implemented a containerized Airflow data platform solution for **multi-cloud deployment,** managing and optimizing **100s of DAGs** (updates, bug fix, migration) across both **managed** (like Composer) and **self-hosted** instances, ensuring robust and reliable workflow automation.
- Built attack path analysis solution using Neo4j Data Science + NetworkX, visualizing exploit chains within complex networks and reducing mean time to remediation from **7 days to 2 days** for critical vulnerabilities.
- Extended support to **7 active product teams** with strict delivery timelines working in **agile model, cross-functional team setup,** showcasing **multitasking ability**, and demonstrated each solution through concise **documentation, flowcharts, diagrams, code reviews** to ensure maintainable code.

- Leveraged GitHub Actions, Azure DevOps, and Terraform to automate and secure the CI/CD pipeline, integrating code quality checks **(SQLFluff, PyRight, PyTest)**, SAST tools **(Mend, SonarQube)**, container security **(Aquasec),** and data quality validation **(Great Expectations, Selenium)**, ensuring secure, and scalable development and deployment processes with a strong focus on **DevSecOps** principles.

Deloitte

Nov 2020 - Apr 2022

Role: Data Engineer **Location: Bangalore, India**

Project Name: K-Governance

Project Description: K-Governance is an enterprise Data Management Platform (DMP) built on Azure, designed to establish a centralized Data Governance framework through a CoE (Center of Excellence). The platform enables effective data stewardship, lineage tracking, and policy enforcement across the organization by forming a unified insights factory and centralizing the storage, management, access, and governance of enterprise data assets to ensure trusted, compliant, and auditable data at scale.

- Designed Azure Data Lake layers based on **medallion architecture** to optimize data storage, retrieval, and enforce **data governance boundaries** across zones with controlled access and ownership policies.
- Developed a robust **metadata-driven ingestion framework** enabling dynamic, configuration-based data onboarding from diverse sources using Azure Data Factory, Azure Functions, Azure Databricks, and Azure SQL; embedded **data quality checks and lineage tracking** as core **governance controls**, with metadata cataloging ensuring full data asset **discoverability and auditability** across the platform.
- Revamped **Informatica ETL workflows** to ensure seamless legacy data migration from on-premises to **Azure Data Lake**, preserving data lineage, schema, and governance metadata throughout the migration lifecycle.
- Implemented end-to-end **CI/CD pipelines** using **Azure DevOps** to accelerate deployment processes; built **Power BI dashboards** for governance SLA tracking, and **FinOps-driven** cost optimization, and cost control to deliver actionable insights for performance tuning and compliance reporting.

Wipro

Aug 2018 – Nov 2020

Role: Data Engineer **Location: Bangalore, India**

Project Name: AI-CoE

Project Description: Addressing client's ambition to extend their AI/ML capabilities in the public sector by forming an AI CoE to promote innovation and accelerate responsible AI adoption.

- Built **batch and real-time ML inference pipelines** using **Azure Databricks**, **Azure ML Service**, and **AKS**, integrating ML predictions into Tableau reports and web applications, with **Denodo acting as** a data virtualization layer between **Oracle Exadata** warehouses and estates.
- Created various Azure Data Factory pipelines for **on-premises to Data Lake ingestion** and **database migration**, ensuring smooth and efficient data transfer with schema validation and error handling.
- Mentored academic and public healthcare sector ML practitioners from the UK on an **MLOps approach and ML lifecycle management at scale**, covering model versioning, drift detection, and retraining strategies.
- Leveraged Azure DevOps to facilitate **CI/CD of ML models** into AKS, monitoring, log collection, and backup patterns to ensure high availability and minimize model deployment downtime.
- **Collaborated closely** with research scientists and healthcare practitioners across Europe to **onboard multiple ML use cases to the AI CoE**, aligning solutions with regulatory and clinical requirements.

SKILLS

Programming & Scripting: Python, PySpark, T-SQL, SQL, Pandas, Unix Scripting, YAML, PowerShell

Cloud Platforms: Microsoft Azure (Data Factory, Data Lake, Synapse, Databricks, Functions, Logic Apps, Purview, Log Analytics, WebApp, AI Foundry, AI Search, OpenAI, ML Services), Google Cloud Platform (BigQuery, Pub/Sub, Cloud Run, Dataproc, Cloud SQL, Cloud Spanner), Microsoft Fabric

Data & Analytics Platforms: Azure Databricks (Autoloader, Delta Live Tables, Structured Streaming, Asset Bundles, Unity Catalog, Genie), Apache Spark, Delta Lake, Azure Synapse, Microsoft Fabric, Google BigQuery

Data Modelling: Relational Data Modelling, Graph Data Modelling, Semantic Layer Modelling, Medallion Architecture, Delta Tables, Star/Snowflake Schema

Data Quality & Governance: Great Expectations, Azure Purview, Unity Catalog, Data Lineage Tracking, Data Classification, Column-Level Security, RBAC, Metadata-Driven Frameworks

Databases: Azure SQL, PostgreSQL, Neo4j, Azure Cosmos DB, Elasticsearch, GCP Cloud SQL, Cloud Spanner

Orchestration & Pipelines: Apache Airflow (Astronomer, GCP Composer, AWS MWAA), Azure Data Factory, DBT, Kafka, Azure Service Bus, Redis

DevOps & CI/CD: Azure DevOps, GitHub Actions, Terraform, JFrog, Docker, Kubernetes (AKS, GKE, EKS), GCP Cloud Build & Cloud Deploy, SonarQube, Mend, Aqua Security

AI/ML Stack: LangChain, LangGraph, AutoGen, Azure AI Foundry, Azure OpenAI, MLflow, KubeFlow, Semantic Kernel, MCP (Model Context Protocol), Pinecone, Azure AI Search, GitHub Copilot, MS365 Copilot Studio

Architecture Patterns: Data Mesh, Data Fabric, AI Fabric, Medallion Architecture, ETL, ELT, DataOps, MLOps, AIOps, Data Governance, Data Quality, Multi-Agent Frameworks, RAG Pipelines

CERTIFICATIONS

- Fabric Data Engineer Associate Certification from Microsoft Inc
- Neo4J Certified Professional from Neo4J Inc
- DataBricks Certified Data Engineer Associate Certification from DataBricks Inc
- DataBricks Certified Gen AI Engineer Associate Certification from DataBricks Inc

EDUCATION

Bachelor of Technology in Information Technology, Sethu Institute of Technology **Aug 2014 – May 2018**